

# MAC21 — Full-Coverage Study Plan

All **35 seven-marker types**, across **both options in every unit** — Q1 and Q2, Q3 and Q4, Q5 and Q6, Q7 and Q8, Q9 and Q10. Plus the 2-mark bank. Nothing skipped.

### The trade-off, stated plainly

Full coverage of 35 types needs about **19 hours**. You have **18**. That is close enough to work, but only because depth is *tiered* rather than coverage being cut.

The honest cost: you get **three reps on the 11 highest-frequency topics** and **one clean rep on everything else**. That is recognition plus one pass, not fluency, on the long tail. It is the right shape for "no question can blindside me" — it is not the shape for "I will finish every question in 20 minutes". You are buying breadth with depth, deliberately.

## Depth key

| LEVEL | TIME   | COUNT | WHAT YOU ACTUALLY DO                                                                                                                                                        |
|-------|--------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DRILL | 38 min | 11    | Read the method, then work <b>three</b> problems — the verbatim repeat plus two others. Third one closed-book, no notes. These are the topics that have appeared 4–6 times. |
| PASS  | 25 min | 17    | Read the method, work <b>one</b> full problem start to finish, read the trap. Do not move on until the worked example comes out right.                                      |
| SKIM  | 13 min | 5     | Read the method and the trap. Write the <b>setup only</b> — first three lines — not the full solve. Enough to start the question in the hall and score method marks.        |

Numbers below match the section numbers in *MAC21-All-7-Markers-Solved.pdf* — so "\$16" means go straight to that section and work it.

## The full inventory — all 35, every section

UNIT I — appears in Q1 and Q22h 44m

| \$ | TOPIC                                               | CAN APPEAR IN | MK  | REPEATS | DEPTH | TIME | REP PLAN                             |
|----|-----------------------------------------------------|---------------|-----|---------|-------|------|--------------------------------------|
| 3  | Taylor expansion, two variables about (1, $\pi/2$ ) | Q2(c)/(d)     | 7   | 4×      | DRILL | 38   | All three variants: cos, sin, $xy^2$ |
| 1  | Newton-Raphson, system                              | Q1(c) / Q2(b) | 7   | 4×      | DRILL | 38   | (0.8,0.4) · (0.8,0.3) · (3.5,−1.8)   |
| 6  | Lagrange multipliers                                | Q1(d) / Q2(d) | 7   | 4×      | PASS  | 25   | Open-top box, then sphere            |
| 5  | Maxima-minima, r-s-t test                           | Q1(d) / Q2(b) | 4–7 | 5×      | PASS  | 25   | $x^3+y^3-3axy$                       |
| 2  | Newton-Raphson, single equation                     | Q2(b)/(c)     | 4–7 | 3×      | PASS  | 25   | Bracket first — see the trap         |
| 4  | Taylor/Maclaurin, one variable                      | Q1(b)         | 4–7 | 5×      | SKIM  | 13   | Setup only                           |

UNIT II — appears in Q3 and Q43h 10m

| \$ | TOPIC                                    | CAN APPEAR IN        | MK  | REPEATS | DEPTH | TIME | REP PLAN                                 |
|----|------------------------------------------|----------------------|-----|---------|-------|------|------------------------------------------|
| 10 | Runge-Kutta 4th order                    | Q3(d) / Q4(d)        | 7   | 6×      | DRILL | 38   | In every paper. Three different ODEs     |
| 9  | Modified Euler's method                  | Q4(c)                | 7   | 5×      | DRILL | 38   | Three, all with two corrector iterations |
| 7  | Taylor's series method                   | Q3(c) / Q4(b)        | 4–7 | 5×      | PASS  | 25   | The $e^{-2t}-2y$ repeat                  |
| 11 | Newton's law of cooling                  | Q3(b)/(c)/(d), Q4(d) | 4–7 | 5×      | PASS  | 25   | Coffee 92/24/80/65                       |
| 12 | Orthogonal trajectories, Cartesian       | Q3(b)/(d), Q4(c)     | 7   | 6×      | PASS  | 25   | Co-axial circles                         |
| 8  | Euler's method                           | Q3(b) / Q4(b)        | 4   | 4×      | SKIM  | 13   | Bungee jumper table                      |
| 13 | Orthogonal traj. polar / self-orthogonal | Q4(c)                | 4–7 | 3×      | SKIM  | 13   | Learn the swap rule only                 |

|    |                  |               |   |    |      |    |                                        |
|----|------------------|---------------|---|----|------|----|----------------------------------------|
| 14 | LR / RC circuits | Q3(d) / Q4(d) | 7 | 3× | SKIM | 13 | Set up the DE, note the resonance case |
|----|------------------|---------------|---|----|------|----|----------------------------------------|

UNIT III — appears in Q5 and Q6
2h 51m

| §  | TOPIC                    | CAN APPEAR IN     | MK  | REPEATS | DEPTH | TIME | REP PLAN                                                  |
|----|--------------------------|-------------------|-----|---------|-------|------|-----------------------------------------------------------|
| 15 | CF + P.I. standard cases | Q5(c) / Q6(c)/(d) | 7   | 6×      | PASS+ | 45   | <b>Foundation — do this first.</b><br>All five P.I. cases |
| 16 | Cauchy's equation        | Q5(c)/(d), Q6(d)  | 7   | 4×      | DRILL | 38   | Three problems, different RHS types                       |
| 18 | Variation of parameters  | Q5(c)/(d), Q6(c)  | 7   | 5×/5    | DRILL | 38   | $\sec^3 x \cdot \tan x \cdot e^3 x/x^2$                   |
| 17 | Legendre's equation      | Q6(c)/(d), Q5(b)  | 4–7 | 4×      | PASS  | 25   | Do straight after Cauchy — same substitution              |
| 19 | IVP / BVP                | Q5(b) / Q6(b)/(d) | 4–7 | 6×      | PASS  | 25   | One IVP, one BVP                                          |

UNIT IV — appears in Q7 and Q8
3h 09m

| §  | TOPIC                                   | CAN APPEAR IN     | MK  | REPEATS | DEPTH | TIME | REP PLAN                                             |
|----|-----------------------------------------|-------------------|-----|---------|-------|------|------------------------------------------------------|
| 22 | Newton's divided difference             | Q7(c)/(d), Q8(c)  | 7   | 5×/5    | DRILL | 38   | The $x^3-2x$ repeat, then two more                   |
| 23 | Missing terms                           | Q8(b)             | 4   | 3×      | DRILL | 38   | The 45...65 table — identical twice                  |
| 26 | Trapezoidal + Simpson 1/3 + Simpson 3/8 | Q7(d), Q8(b)/(d)  | 4–7 | 5×      | PASS  | 25   | <b>All three on one integral</b> — $\int dx/(1+x^2)$ |
| 20 | Newton-Gregory forward / backward       | Q7(c)             | 7   | 2×      | PASS  | 25   | Overdue — f(14.2) backward                           |
| 21 | Lagrange interpolation + inverse        | Q7(b)/(c), Q8(c)  | 4–7 | 3×      | PASS  | 25   | Forward, then the inverse form                       |
| 24 | Grouped-frequency estimation            | Q7(c) / Q8(c)/(d) | 7   | 3×      | PASS  | 25   | The cumulative-column trick                          |
| 25 | Numerical differentiation               | Q7(d) / Q8(c)     | 7   | 4×      | SKIM  | 13   | Both formulas written out, one setup                 |

UNIT V — appears in Q9 and Q10
3h 34m

| §  | TOPIC                           | CAN APPEAR IN      | MK  | REPEATS    | DEPTH | TIME | REP PLAN                                        |
|----|---------------------------------|--------------------|-----|------------|-------|------|-------------------------------------------------|
| 32 | Eigenvalues & eigenvectors      | Q9(b)/(c)/(d)      | 4–7 | foundation | DRILL | 38   | <b>Unlocks \$33, \$34, \$35.</b> Do 2×2 and 3×3 |
| 35 | System of ODEs by matrix method | Q10(d)             | 7   | 3×/3       | DRILL | 38   | Final question in all 3 papers                  |
| 31 | Gauss-Seidel                    | Q9(b) / Q10(b)/(c) | 4–7 | 3×/3       | DRILL | 38   | Reorder first — see the trap                    |
| 33 | Diagonalization + $A^n$         | Q9(c)/(d)          | 7   | 3×/3       | PASS  | 25   | Straight after §32                              |
| 34 | Rayleigh's power method         | Q9(c)/(d), Q10(b)  | 4–7 | 3×/3       | PASS  | 25   | Note the slow-convergence trap                  |
| 30 | Gauss elimination, word problem | Q9(d) / Q10(c)     | 7   | 2×         | PASS  | 25   | Dietician; write the system first               |
| 29 | Rank, echelon & consistency     | Q9(b) / Q10(c)     | 4–7 | 2×         | PASS  | 25   | Condition + parametric family                   |

SUNDAY 15:00 → 00:00
(5.5 h)

| TIME | BLOCK | SECTIONS, IN ORDER |
|------|-------|--------------------|
|------|-------|--------------------|

|             |                                                   |                                                                                                                                                                                                                                                                                                                                    |
|-------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:00–18:00 | <b>UNIT III — complete</b><br><b>Q5 and Q6</b>    | <b>§15 CF + P.I. 45m</b> — all five P.I. cases. Everything else in this unit rests on it, so do not rush it.<br><b>§16 Cauchy</b> <b>DRILL</b> 38m<br><b>§17 Legendre 25m</b> — immediately after Cauchy, same substitution with an extra $a^2$<br><b>§18 Variation of parameters</b> <b>DRILL</b> 38m<br><b>§19 IVP / BVP 25m</b> |
| 18:00–21:30 | Break                                             | Long one. Eat properly — there is a late block coming.                                                                                                                                                                                                                                                                             |
| 21:30–00:00 | <b>UNIT V — the eigen chain</b><br><b>Q9 side</b> | <b>§32 Eigenvalues &amp; eigenvectors</b> <b>DRILL</b> 38m — a $2 \times 2$ and a $3 \times 3$<br><b>§33 Diagonalization + <math>A^n</math> 25m</b><br><b>§34 Rayleigh's power method 25m</b><br><b>§35 System of ODEs by matrix</b> <b>DRILL</b> 38m<br>Then 20m: Unit III + Unit V definitions, written from memory.             |
| 00:00–07:00 | Sleep                                             | 7 hours.                                                                                                                                                                                                                                                                                                                           |

**End of Sunday:** Unit III complete (both Q5 and Q6), and the whole eigenvalue chain — four of Unit V's seven types, including the two that are  $3 \times 3$ .

## MONDAY 10:30 → 22:45 (~10.5 h)

| TIME        | BLOCK                                                   | SECTIONS, IN ORDER                                                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07:00–10:30 | Wake, workout, eat                                      | Arrive at the desk fed.                                                                                                                                                                                                                                                                                    |
| 10:30–13:00 | <b>UNIT V finish + UNIT IV start</b><br><b>Q10 side</b> | <b>§31 Gauss-Seidel</b> <b>DRILL</b> 38m — reorder for dominance first<br><b>§30 Gauss elimination 25m</b><br><b>§29 Rank &amp; consistency 25m</b> → Unit V now complete, both sides<br><b>§23 Missing terms</b> <b>DRILL</b> 38m<br><b>§26 Trapezoidal + both Simpson rules 25m</b>                      |
| 13:00–13:45 | Lunch                                                   | —                                                                                                                                                                                                                                                                                                          |
| 13:45–16:00 | <b>UNIT IV — finish</b><br><b>Q7 and Q8</b>             | <b>§22 Divided difference</b> <b>DRILL</b> 38m<br><b>§20 Newton-Gregory fwd/bwd 25m</b><br><b>§21 Lagrange interpolation + inverse 25m</b><br><b>§24 Grouped frequency 25m</b><br><b>§25 Numerical differentiation</b> <b>SKIM</b> 13m                                                                     |
| 16:00–16:20 | Break                                                   | Compulsory.                                                                                                                                                                                                                                                                                                |
| 16:20–18:20 | <b>UNIT II — numerical methods</b><br><b>Q3 and Q4</b>  | Learn these four as <i>one family</i> — same skeleton, different weighting.<br><b>§9 Modified Euler</b> <b>DRILL</b> 38m<br><b>§10 Runge-Kutta 4th order</b> <b>DRILL</b> 38m<br><b>§7 Taylor's series method 25m</b><br><b>§8 Euler's method</b> <b>SKIM</b> 13m                                          |
| 18:20–19:00 | Dinner                                                  | Compulsory.                                                                                                                                                                                                                                                                                                |
| 19:00–20:15 | <b>UNIT II — applications</b>                           | <b>§11 Newton's law of cooling 25m</b><br><b>§12 Orthogonal trajectories, Cartesian 25m</b><br><b>§13 Polar / self-orthogonal</b> <b>SKIM</b> 13m<br><b>§14 LR / RC circuits</b> <b>SKIM</b> 13m → Unit II complete                                                                                        |
| 20:15–22:45 | <b>UNIT I — complete</b><br><b>Q1 and Q2</b>            | <b>§3 Two-variable expansion</b> <b>DRILL</b> 38m — all three variants<br><b>§1 Newton-Raphson system</b> <b>DRILL</b> 38m<br><b>§6 Lagrange multipliers 25m</b><br><b>§5 Maxima-minima 25m</b><br><b>§2 Newton-Raphson single 25m</b> — bracket first<br><b>§4 One-variable expansion</b> <b>SKIM</b> 13m |
| 22:45–23:15 | Wind down                                               | Sleep by 23:15.                                                                                                                                                                                                                                                                                            |

**End of Monday:** all 35 types covered, both sides of all five units.

## TUESDAY 06:15 → 08:50

| TIME        | BLOCK              | WHAT EXACTLY                                                                                                                                                                                                                                                                                   |
|-------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06:15–06:50 | Wake, breakfast    | Eat properly. Do not skip.                                                                                                                                                                                                                                                                     |
| 06:50–07:35 | 2-mark bank        | Every definition, all five units, written from memory. 10 marks and the cheapest of the paper.                                                                                                                                                                                                 |
| 07:35–08:35 | SPEED RUN — all 35 | Timed. For each of the 35 sections write <b>only the first three lines</b> — the setup, not the solve. <b>100 seconds each</b> . Anything you cannot start in 100 seconds, mark it and re-read that one method box immediately. This is your timing rehearsal and your gap-finder in one hour. |
| 08:35–08:50 | Traps only         | Read the red TRAP boxes end to end. Nothing else.                                                                                                                                                                                                                                              |
| 08:50       | STOP               | Travel. Learn nothing new.                                                                                                                                                                                                                                                                     |

## If you fall behind

**Compress in this order — never drop a topic entirely, downgrade its depth instead.** Coverage is the whole point of this version of the plan; a topic at **SKIM** still lets you start the question and take method marks.

1. Downgrade these four from **PASS** to **SKIM** (saves 48 min): §20 Newton-Gregory · §21 Lagrange interpolation · §29 rank & consistency · §30 Gauss elimination.
2. Downgrade §17 Legendre and §19 IVP/BVP to **SKIM** (saves 24 min) — both are short riders on work you have already done.
3. Cut the **DRILL** topics from three reps to two (saves ~50 min across the eleven). Do this last — those are the highest-frequency questions in the paper.

**Never compress:** §15 CF + P.I. (everything in Unit III depends on it) · §32 eigenvalues (§33, §34, §35 all depend on it) · the 2-mark bank · Tuesday's speed run.

## In the hall — 09:30 to 12:30

| MINUTES | DO THIS                                                                                                                                                                                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–5     | <b>Read all ten questions.</b> With full coverage you can genuinely choose — in each unit pick the side where you can start <i>more parts</i> , not the one that looks prettier. Write your five choices at the top. |
| 5–20    | <b>All five 2-markers.</b> 10 marks in 15 minutes.                                                                                                                                                                   |
| 20–50   | <b>All five 4-markers.</b> 30 marks banked in 50 minutes.                                                                                                                                                            |
| 50–170  | <b>The ten 7-markers, easiest first.</b> 12 minutes each. You have prepared every one of them — so lead with the units you drilled. If one stalls past 15 minutes, leave it and return.                              |
| 170–180 | <b>Sweep.</b> Check numbering. Fill any blank with formula + setup — 2–3 marks for ninety seconds.                                                                                                                   |

**Because you covered both sides, attempt both 7-markers wherever you have time.** That is the payoff for the breadth: the plan's floor is no longer 13 marks a unit, it is whatever you can write in the time. Show every table — these are method-marked papers.

All 35 seven-marker types across both options in every unit, plus the 2-mark bank. Section numbers match *MAC21-All-7-Markers-Solved.pdf*. Repeat counts are out of the papers carrying each unit (I–III: 6 · IV: 5 · V: 3). Budget is about 19 hours of work against roughly 18 available, which is why depth is tiered; the compression order above is the intended release valve. Timings assume a 15:00 Sunday start and sleep by 23:15 both nights.